

IBONG TIRIRIT (PIPE 4)
IPD PPD REFRESHERS TRIVIA EXAMINATION
March 2010

MULTIPLE CHOICE QUESTIONS

Select the best answer from each of the following questions. On the answer sheet provided, shade the box that corresponds to your choice. Strictly no erasures allowed.

1. An air-vapor mixture has a DB temperature of 30 °C and a humidity ratio of 0.015 kg/kg d.a. Calculate the enthalpy.
a) **68.527 kJ/kg d.a.** b) 78.527 kJ/kg d.a. c) 58.527 kJ/kg d.a. d) 88.527 kJ/kg d.a.
2. What is the specific gravity of a fuel at 28° Baume?
a) 0.868 0.688 **c) 0.886** d) 0.986
3. Calculate the moisture content of moist air at 20 °C DB, 15 °C WB, and 95 kPa barometric pressure. Let the saturation pressure of water at 15 °C equals to 1.7057 kPaa and at 20 °C is 2.339 kPaa.
a) **0.00923 kg/kg d.a.** b) 0.00293 kg/kg d.a. c) 0.00392 kg/kg d.a. d) 0.00239 kg/kg d.a.
4. Determine the mass of a nitrogen gas confined in a 6 m³ tank at a pressure of 198.675 kPag and temperature of 40 °C.
a) 18.4 kg **b) 19.4 kg** c) 20.4 kg d) 17.4 kg
5. The heat rejected from condenser is 300 kW. The water required to cool the refrigerant is 5 lps. Determine the temperature of water leaving the condenser if water enters at 25 C.
a) **39.33 °C** b) 33.39 °C c) 33.93 °C d) 43.93 °C
6. Determine the resulting pressure when one kilogram of air at 103 kPaa and 95 °C is heated at constant volume to 425 °C.
a) 205.36 kPaa **b) 195.35 kPaa** c) 196.35 kPaa d) 193.56 kPaa
7. What is the amount of heat added to 15 kg of air to increase its temperature from 30 °C to 250 °C without changing the pressure?
a) 3230.46 kJ b) 3420.26 kJ **c) 3320.46 kJ** d) 3640.36 kJ
8. Determine the compressor work of a Brayton cycle if its pressure ratio is 8 and the initial temperature is 32 °C.
a) **248.67 kJ/kg** b) 278.47 kJ/kg c) 284.67 kJ/kg d) 267.84 kJ/kg
9. If the specific humidity of moist air is 0.027 at 1 atmosphere pressure. What is the partial pressure of the water vapor in the mixture?
a) **4.215 Kpaa** b) 3.516 Kpaa c) 3.617 Kpaa d) 2.584 kPaa
10. If the enthalpy of a certain gas is 2400 kJ/kg and its internal energy is 2200 kJ/kg, find the specific volume at a pressure of 950 kPaa.
a) **0.2105 m³/kg** b) 0.2501 m³/kg c) 0.2015 m³/kg d) 0.1025 m³/kg
11. Determine the maximum temperature of a Carnot cycle if it operates at a minimum temperature of 20 °C with an efficiency of 60 %.
a) **459.5 °C** b) 455.9 °C c) 495.5 °C d) 559.4 °C
12. Air at 1 MPaa and 310 °C is expanded to 250 kPaa in accordance with the relation $pV^{1.25} = C$. Determine the work done.
a) **+ 162.11 kJ/kg** b) - 162.11 kJ/kg c) + 126.11 kJ/kg d) - 126.11 kJ/kg
13. A Rankine cycle operates with a thermal efficiency of 40 % and the factor of evaporation of the boiler is 1.15. Determine the mass flow rate of steam if the cycle power output is 5.5 MW.
a) **5.3 kg/s** b) 4.3 kg/s c) 3.5 kg/s d) 6.3 kg/s
14. The mixing ratio of humid air is 0.034 at a dry-bulb temperature of 27°C. Determine the enthalpy of the given air. $h_g @ 27 °C = 2550.8$ KJ/kg.
a) 67.25 kJ/kg b) 80.62 KJ/kg **c) 103.69 KJ/kg** d) 55.68 KJ/kg
15. The dry bulb temperature of humid air is 35°C. What is the wet-bulb depression in °F if the wet-bulb temperature is 18°C?
a) 17.6 **b) 30.6** c) 18.0 d) 25.8
16. The enthalpies at the entrance and exit of a steam engine are 2700 kJ/kg and 2000 kJ/kg, respectively. The steam consumption of the engine is 1000 kg/hr. If the indicated efficiency is 66 % and the mechanical efficiency is 92 %, determine the engine brake power output.
a) **118.06 kW** b) 181.06 kW c) 110.86 kW d) 168.01 kW
17. The percent rating of water tube boiler is 200 %, factor of evaporation is 1.10, and heating surface is 400 ft². Determine the rate of evaporation, in kg/hr.

- a) 1831 b) 1831 **c) 1138** d) 1813
18. A boiler operates at 82.5 % efficiency while the mass of steam generated is 408 200 kg in 5 hours. The enthalpy of steam is 3187 kJ/kg and feed is 604.83 kJ/kg while the fuel used for boiler has a heating value of 32 567.85 kJ/kg. Find the mass of fuel needed per day in metric tons.
a) 188.3 b) 183.8 c) 138.8 d) 318.8
19. A 10 kg/s of air enters the theater at 16 °C. The theater is to be maintained at 27 °C DB and 20 °C WB. If the sensible heat ratio is 0.80, what is the latent heat load of the theater?
a) 27.67 kW b) 26.75 kW c) 110.682 kW d) 72.67 kW
20. Determine the mole fraction of the helium in a mixture of 45 % helium and 55 % nitrogen by mass analysis.
a) 14.84 % **b) 85.16 %** c) 18.44 % d) 81.56 %
21. A 60 MW power plant has an average load of 34 500 kW and a load factor of 0.75. Find the reserve power over a peak load power.
a) 14 000 kW b) 20 000 kW c) 46 000 kW d) 5 000 kW
22. A steam generator burns fuel oil with 20 % excess air. Fuel may be represented by $C_{14}H_{30}$. Calculate the actual-air fuel ratio.
a) 17.9 b) 18.7 c) 16.7 d) 15.7
23. A certain coal has the following ultimate analysis by weight. C = 67 %, Ash = 5 %, Moisture = 8 %, N = 6 %, H = 3 %, Sulfur = 7 %, O = 4 %. Calculate the air-fuel ratio if this coal is burned with 50 % excess air.
a) 12.4 **b) 13.3** c) 14.3 d) 12.4
24. A container filled with helium is dropped 4 km above the ground, find the change of temperature?
a) 12.6 °C b) 17.5 c) 15.2 °C d) 18.5 °C
25. 2.5 m³/s of humid air at a state of 28°C DB, 21 WB and 101.325 Kpa flow across a cooler coil and leaves at 12.5°C DB and 0.0083 kg vapor per kg dry air. If the apparatus dew point (ADP) is 8.5, compute for the contact factor (CF).
a) 0.835 **b. 0.795** c. 0.579 d. 0.306
26. A pump draws water at a head of 110 m and a discharge rate of 550 lps. Determine the brake power input if the pump efficiency is 70 %.
a) 847.57 kW b) 874.57 kW c) 875.74 kW d) 854.75 kW
27. A turbine has a peripheral coefficient of 0.6. Find the runner diameter of the turbine if it operates at 450 rpm and a head of 60 m.
a) 0.874 m b) 0.784 m c) 0.478 m d) 0.748 m
28. A 5 m x 5 m wall of a room has a surface temperature of 40 °C. The wall thickness is 25 cm and has a thermal conductivity of 0.07 W/m-°K, and air film conductance is 15 W/m²-°K. If the outside air temperature is 18 °C, what is the heat loss through the wall?
a) 151.18 W b) 185.18 W c) 158.11 W d) 118.15 W
29. An air compressor operates with discharge pressure of 6 times the suction pressure. If the volume flow rate at the suction is 0.8 m³/s and the compression process is in accordance with $pV^{1.38} = C$, determine the compressor power, in kW.
a) 187.16 kW b) 178.16 kW c) 167.18 kW d) 168.17 kW
30. A horizontal pipe carries oil with a specific gravity of 0.83. If two pressure gauges along the pipe read 515 kPa and 430 kPa, respectively. Determine the head loss between the two gauges.
a) 10.44 m b) 14.40 m c) 14.04 m d) 40.14 m
31. Air at 33 °C DB and 19 °C WB is cooled and humidified by passing through an air washer in which water is continuously recirculated. The air leaves the air washer at 23 °C DB. What is the efficiency of the air washer?
a) 71.43 % b) 73.41 % c) 74.31 % d) 73.14 %
32. A stream of outdoor air is mixed with a stream of return air in an air conditioning system that operated at 101.325 kPa barometric pressure. The flow rate of outdoor air is 2 kg/s, and its conditions are 35 °C DB, 25 °C WB, and 0.0159 kg/kg d.a.. The flow rate of return air is 3 kg/s, and its conditions are 24 °C DB, 50 % RH, and 0.00925 kg/kg d.a. Determine the temperature of the mixture.
a) 28.4 °C b) 26.4 °C c) 29.4 °C d) 30.4 °C
33. Determine the indicated power of a four-cylinder, 4-stroke, Diesel engine with 20-cm bore and 30-cm stroke running at 1000 rpm and has a reading of 450 kPa mean effective pressure in the indicator diagram.
a) 159.83 Hp **b) 189.53 Hp** c) 158.93 Hp d) 198.53 Hp
34. The approach and efficiency of cooling tower are 10 °C and 65 %, respectively. If the temperature of water leaving the tower is 27 °C, determine the temperature of water entering the tower.
a) 54.57 °C b) 55.47 °C **c) 45.57 °C** d) 54.75 °C

35. If the sensible heat ratio is 0.80 and the cooling load is 100 kW, what is the amount of sensible heat?
a) 100 kW b) 125 kW **c) 80 kW** d) 60 kW
36. A dryer is to deliver 1000 kg/hr of cassava with 2 % moisture and 20 % moisture in the feed. Determine the mass of air required per hour if the change in humidity ratio is 0.0165.
a) 16 136.36 kg/hr **b) 13 636.36 kg/hr** c) 13 163.63 kg/hr d) 16 336.63 kg/hr
37. The change of enthalpy in an air conditioning unit is 23.5 kJ/kg. The mass of supply air is 68000 kg/hr. What is the conditioner capacity, in TR?
a) 126.21 TR b) 125.26 TR c) 162.25 TR d) 152.26 TR
38. A DC motor driven pump running at 100 rpm delivers 30 lps of water against a total pumping head of 27 m with a pump efficiency of 60 %. What speed would result if the pump rpm is increased to a pumping head of 36 m assuming no change in efficiency?
a) 115.47 rpm b) 114.57 rpm c) 117.54 rpm d) 114.75 rpm
39. A mechanical draft cooling tower cools the cooling water from 60°C to 25°C at the rate of 8 kg/sec. Atmospheric air enters the tower at a state of 20°C DB and 16°C WB. The air leaves the tower at 35°C. What is the temperature of approach?
a. 9°C b. 6°C c. 8°C d. 35°C
40. In a geothermal power plant, the enthalpies of the ground water and the turbine inlet are 1500 kJ/kg and 3500 kJ/kg respectively. If the enthalpy of the hot water in the flash tank is 700 kJ/kg and the mass flow rate of steam is 15 kg/s, what is the mass flow rate of ground water?
a) 52.5 kg/s b) 55.2 kg/s c) 25.5 kg/s d) 35.2 kg/s
41. 12 kg of gas with a molecular mass of 30 at 101.325 kPa pressure and 30 °C temperature occupies a volume of 9500 liters. Determine the compressibility factor.
a) 0.955 b) 0.595 c) 0.795 d) 0.895
42. A refrigeration compressor has a specific volume of 0.0482 m³/kg at entrance and 0.017 m³/kg at the exit. If volumetric efficiency is 90 %, determine the percent clearance of the compressor.
a) 5.45 % b) 5.54 % c) 4.55 % d) 5.65 %
43. In an air-standard Brayton cycle, air enters compressor at 1 bar and 15 °C. The pressure leaving the compressor is 0.6 MPaa and maximum temperature of the cycle is 1000 °C. Calculate the cycle thermal efficiency.
a) 35 % b) 32 % c) 37.5 % **d) 40 %**
44. Liquid ammonia at a temperature of 26 °C is available at the expansion valve. The temperature of the vaporizing ammonia in the evaporator is 2 °C. Find the percentage of liquid vaporized while flowing through the expansion valve. Properties of refrigerant: At t = 2 °C, h_f = 190.4 kJ/kg and h_g = 1445.6 kJ/kg; At 26 °C, h_f = 303.6 kJ/kg and h_g = 1465.6 kJ/kg.
a) 10.2 % **b) 9.02 %** c) 11.02 % d) 12.02 %
45. An ideal dual combustion cycle operates on 0.45 kg of air. At the beginning of compression, the air is 97 kPa, 43 °C. Let r_p = 1.5, r_c = 1.6, and r_k = 11. Determine the percent clearance.
a) 10 % b) 11 % c) 9 % d) 12 %
46. An ideal dual combustion cycle operates on 0.45 kg of air. At the beginning of compression, the air is 97 kPa, 43 °C. Let r_p = 1.5, r_c = 1.6, and r_k = 11. Determine the thermal efficiency.
a) 58.7 % b) 57.8 % c) 48.7 % d) 68.7 %
47. In an air standard Otto cycle, the clearance volume is 18 % of the displacement volume. Find the thermal efficiency. (10/93)
a) 52 % **b) 53 %** c) 55 % d) 60 %
48. Helium is used in a Carnot engine where the volumes beginning with the constant temperature heat addition are V₁ = 0.3565 m³, V₂ = 0.5130 m³, V₃ = 8.0 m³, and V₄ = 5.57 m³. Determine the thermal efficiency if k = 1.666.
a) 83.95 % b) 89.35 % c) 85.93 % d) 85.39 %
49. Determine the equivalence ratio of the following mixture: 4 grams of butane, C₄H₁₀, and 75 grams of air.
a) 0.72 b) 0.92 c) 0.62 **d) 0.82**
50. A hydraulic turbine receives water from a reservoir at an elevation of 100 m above it. What is the minimum water flow in kg/s to produce a steady turbine output of 50 MW?
a) 51 000 b) 15 000 c) 50 100 d) 50 010
51. An ideal vapor compression cycle requires 2.5 kW to power the compressor. You have found the following data for the cycle: the enthalpy at the condenser entrance = 203 kJ/kg, exit = 55 kJ/kg; and evaporator entrance = 55 kJ/kg, exit = 178 kJ/kg. If the mass flow rate of the refrigerant is 0.10 kg/s, what is the coefficient of performance of the system?

- a) 9.42 **b) 4.92** c) 4.29 d) 2.94
52. A 100 TR system is used to produce chilled water from 22 °C to 2 °C. Determine the volume flow rate of water in lps.
a) 3.2 lps **b) 4.2 lps** c) 3.8 lps d) 4.8 lps
53. A refrigeration system operates on the reversed Carnot cycle. The minimum and maximum temperatures are – 25 °C and 72 °C, respectively. If the heat rejected at the condenser is 6000 kJ/min, determine the power input required.
a) 1686.83 kJ/min b) 1668.83 kJ/min c) 1866.83 kJ/min d) 1683.68 kJ/min
54. Fish weighing 11 000 kg with a temperature of 20 °C is brought to a cold storage and which shall be cooled to – 10 °C in 11 hours. Find the required plant refrigerating capacity in TR if the specific heat of fish is 0.7 kCal/kg-°C above freezing and 0.30 kCal/kg-°C below freezing point that is – 3 °C. The latent heat of freezing is 55.5 kCal/kg.
a) 25.26 b) 14.38 c) 15.26 **d) 24.38**
55. Determine the brake power of the engine having a brake thermal efficiency of 35 % and uses 25 °API fuel with fuel consumption of 40 kg/hr.
a) 160.67 kW **b) 173.52 kW** c) 174.52 kW d) 165.84 kW
56. A refrigeration compressor has a specific volume of 0.0482 m³/kg at entrance and 0.017 m³/kg at the exit. If volumetric efficiency is 90 %, determine the percent clearance of the compressor.
a) 5.45 % b) 5.54 % c) 4.55 % d) 5.65 %
57. In a Carnot cycle, heat addition occurs during which of the following processes?
a) Isothermal compression process c) Isentropic expansion process
c) Isothermal expansion process d) isentropic compression process
58. Stirling cycle is also called as:
a) Constant volume regenerative cycle b) Constant pressure regenerative cycle
c) Constant temperature regenerative cycle d) Constant entropy regenerative cycle
59. Ericson cycle is also called as:
a) Constant volume regenerative cycle **b) Constant pressure regenerative cycle**
c) Constant temperature regenerative cycle d) Constant entropy regenerative cycle
60. It is the generic term given to a family of solid fuels with high carbon content. How do you call this?
a) Solid fuel **b) Coal** c) Anthracite d) Bituminous
61. Pressure drop caused by friction in fully developed fluid flow can be determined directly using which of the following?
a) Computational fluid dynamics b) The Bernoulli's equation
c) The Darcy-Weisbach equation d) The first law of thermodynamics
62. The existing 2000 TR chiller at a large hospital, where you are assigned as Engineer, must be replaced. Which of the following types most likely to be installed?
a) Screw chiller b) Scroll Chiller c) Reciprocating chiller **d) Centrifugal chiller**
63. Which of the following parameters has the greatest limiting effect on the thermal performance of an open, evaporative cooling tower?
a) Wet-Bulb temperature b) Dry-Bulb temperature c) Range c) Approach
- Note: It is not possible to cool the water below the air's wet-bulb temperature. Thus, the WB temperature, more than any other parameter, sets the absolute limit on the cooling tower's capacity.
64. The amount of heat required melting a ton (2000 lbs) of ice at 32°F, 288,000 Btu/24 hr, 12,000 Btu/hr, or 200 Btu/min.
a) KW of Refrigeration **b) Tons of Refrigeration** c) Refrigerating Effect d) Latent Heat
65. It is the temperature of vapor refrigerant above its saturation temperature corresponding to the given pressure..
a) Saturated **b) Superheat** c) Subcooled d) Absolute
66. What is the chemical name of R-23?
a) Trifluoromethane b) Dichlorofluoromethane c) Dichlorotetrafluoroethane d) Trichlorofluoromethane
67. In a vapor-compression refrigeration system, evaporator is a component normally called as:
a) High-pressure vapor heat exchanger **b) Low-pressure heat exchanger**
c) Throttling device d) Pressure-raising device
68. It is a component in the refrigeration system that returns the lubricating oil, carried over by the refrigerant vapors, to the compressor. How do you call this component?
a) Liquid receiver **b) Oil separator** c) Accumulator d) Capillary tube
69. It is a boiler fuel created from the residue produced from crude petroleum after it has been distilled to produce lighter oils like gasoline, paraffin, kerosene, diesel or gas oil. What is this fuel?
a) Oil b) Kerosene c) Coke d) Peat
70. What do you call a form of boiler fuel that is easy to burn, with very little excess air?

- a) Coal b) Oil c) **Gas** d) Bunker
71. It defined as energy added to or removed from a system divided by the mean absolute temperature over which the change takes place. How do you call this?
- a) Enthalpy b) Heat energy c) Specific heat d) **Entropy**
72. The second law of thermodynamics says that whenever energy is exchanged or converted from one form to another, the potential for energy to do work will be _____.
a) **Decreased** b) Increased c) Zero d) Stagnant
73. It is a space, or several rooms, or units of space having some sort of coincident loads or similar operating characteristics. It may or may not be an enclosed space, or it may consist of many partitioned rooms. How do you call this?
a) **Zone** b) Room c) Space d) Plenum
74. In air conditioning, it indicates either a volume or a site without a partition or a partitioned room or group of rooms. How do you call this?
a) Zone b) Room c) **Space** d) Plenum
75. What is true about steam as it condenses?
a) It does so at constant entropy and temperature
b) It does so at constant enthalpy and reducing temperature
c) **Both enthalpy and entropy reduce and temperature remains constant**
d) Both enthalpy and entropy increase
76. It is an increase in fluid pressure in a long pipe caused by a sudden velocity decrease, where the sudden decrease in velocity is caused by a valve closing.
a) Aero dynamics b) **Water hammer** c) Terminal velocity d) Drag
77. It is the term referring to any glazed aperture in a building envelope; the components of which include the following: glazing material, either glass or plastic; framing, mullions, muntins, and dividers; external shading devices; and integral-shading systems (between glass). What is this?
a) Infiltration b) Radiation c) **Fenestration** d) Transmission
78. It is an envelope area with significantly higher rate of heat transfer than the contiguous enclosure. How do you call this?
a) **Thermal Bridge** b) Thermal Radiation c) Thermal Surface d) Thermal area
79. How is the water in a pressurized deaerator heated to the required temperature?
a) By a blanket of steam above the water b) By direct steam injection into the water
c) **By a spray of steam as it enters the deaerator dome** d) It is not heated further but just held at a higher pressure
80. It is the rate, in Btu/h or W, of heat transfer at the coil. Its cooling load component is the rate at which heat is removed by the chilled water flowing through the coil or is absorbed by the refrigerant inside the coil. What do you call this?
a) **Coil load** b) Heating coil load c) Refrigerating load d) Cooling load
81. It is the rate, in Btu/hr or W, at which heat is added to the conditioned air from the hot water, steam, or electric heating elements inside the coil.
a) Coil load b) **Heating coil load** c) Refrigerating load d) Cooling load
82. It is an upward force that is exerted on an object (like flat plate, airfoil, rotating cylinder, etc.) as the object passes through a fluid. What do you call this?
a) Drag b) **Lift** c) Aero horsepower d) Terminal velocity
83. What is a frictional force that acts parallel but opposite to the direction of motion of fluid?
a) **Drag** b) Lift c) Aero horsepower d) Terminal velocity
84. It is a humidity sensor used to measure relative humidity, dew point, or absolute humidity of ambient or moving air. This device could be mechanical or electronic. How do you call this?
a) Hydrometer b) **Hygrometer** c) Psychrometer d) Barometer
85. It is a device designed to control the flow of steam, water, gas, or other fluids. It can be considered a variable orifice positioned by an actuator in response to impulses or signals from the controller. It may be equipped with either a throttling plug, V-port, or rotating ball specially designed to provide a desired flow characteristic. How do you call this device.
a) **Automatic Valve** b) Automatic flow meter c) Thermostat d) Pyrometer
86. It is a term used by automobile manufacturers to designate the power required to move a car horizontally at 50 mi/hr against the drag force. What is this?
a) Mechanical horsepower b) **Aero horsepower** c) Fuel power d) Fluid power
87. A manufacturer claims that it has a COP of 2.0 when cooling a food at 5 °C using ambient air at 30 °C as a heat sink. Is the claim of the manufacturer valid?
a) **Yes** b) No c) Probably d) It needs experimental verification
88. A storage chamber for low-side liquid refrigerant; also known as surge drum, or surge header. Also, a pressure vessel whose volume is used in a refrigerant circuit to reduce pulsation.
a) Receiver b) **Accumulator** c) Absorber d) Oil separator

89. A form of aluminum oxide that absorbs moisture readily and is used as a drying agent.
a) Calcium chloride b) Lithium bromide c) Activated carbon **d) Activated alumina**
90. A form of carbon made porous by special treatment by which it is capable of absorbing various odors, anesthetics and other vapors.
a) Activated carbon b) Activated alumina c) Absorber d) Absorbent
91. What can you say about the velocity of fluid at the center of the pipe section?
a) Maximum b) Minimum c) Average d) Zero
92. Continuity equation is applicable to which of the following?
a) Viscous and unviscous fluids b) Compressibility fluids **c) Conservation of mass** d) Steady and unsteady flows
93. Hygrometer is an instrument used to determine of which of the following?
a) Specific gravity of fluids b) Specific gravity of gases c) Specific gravity of liquids d) Specific gravity of solids
94. The temperature at which the condensation of water vapor in a space begins for a given state of humidity and pressure as the temperature of the vapor is reduced. This is the temperature corresponding to saturation for a given absolute humidity at constant pressure.
a) Dew point temperature b) Wet bulb temperature c) Dry bulb temperature d) Saturation temperature
95. The temperature of gas or mixture of gases indicated by an accurate thermometer after correction for radiation.
a) Dew point temperature b) Wet bulb temperature **c) Dry bulb temperature** d) Saturation temperature
96. The temperature registered by a thermometer whose bulb is covered by a wetted wick and exposed to a current of rapidly moving air.
a) Dew point temperature **b) Wet bulb temperature** c) Dry bulb temperature d) Saturation temperature
97. The ratio of the actual water vapor pressure of the air to the saturated water vapor pressure of the air at the same temperature is known as:
a) Relative humidity b) Humidity ratio c) Specific humidity d) Percent humidity
98. The weight of water vapor in grains, pounds, grams, or kilograms per grain, pound, gram, or kilogram of dry air.
a) Relative humidity b) Specific volume **c) Specific humidity** d) Percent humidity
99. At critical point of a pure substance, what is the value of latent heat?
a) Maximum b) Minimum **c) Zero** d) Below zero
100. Which of the following is a function of a steam nozzle?
a) Converts kinetic energy into heat energy b) Changes internal energy into kinetic energy
c) Converts potential energy into heat energy **d) Changes enthalpic energy into kinetic energy**
101. If the sensible heat ratio is 0.75, what does it mean?
a) 75 % latent heat and 25 % sensible heat **b) 75 % sensible heat and 25 % latent heat**
c) 25 % latent and sensible heat and 75 % latent heat d) 75 % sensible and latent heat and 25 % sensible heat
102. The hardness of ground water required as feed water for a boiler is 0 – 10 ppm (part per million), which of the following is the range of alkalinity requirement?
a) pH 9 to pH 10 b) pH 6 to pH 10 **c) pH 10 to pH 11** d) pH 8 to pH 10
103. As used in the Revised PSME Code for Boilers and Pressure Vessels, it is any boiler which does not exceed any of the following limits: 405 mm inside diameter, 1065 mm overall length of outside to outside heats at center, 1.85 m² of water heating surface, and 7.03 kg/cm² maximum allowable working pressure. What is this boiler?
a) Power boiler b) Portable boiler **c) Miniature boiler** d) Locomotive boiler
104. It is a closed vessel in which steam or other vapor (to be used externally to itself) is generated by the direct application of heat used for power generation. How do you call this boiler?
a) Power boiler b) Portable boiler c) Oil-fired boiler d) Miniature boiler
105. Name one characteristic feature of **mechanical** steam traps.
a) They pass condensate at steam temperature b) They operate by sensing condensate temperature
c) They can be fitted into any position d) They are not affected by increasing back pressure
106. In a mechanical steam trap, why is a float trap better at venting air than an inverted bucket trap?
a) A float can quickly adjust to the presence of air **b) A float is fitted with an automatic air vent**
c) A float trap does not vent air better than bucket trap d) The air vent orifice is adjustable on a float trap
107. The thermal efficiency of Stirling cycle is the same as the efficiency of which of the following cycles?
a) Otto cycle b) Brayton cycle **c) Carnot cycle** d) Dual cycle
108. In a Brayton cycle or gas turbine cycle, what do you call the heat exchanger that passes exhaust heat to the compressed air, prior to combustion?
a) Reheater **b) Recuperator** c) Heater d) Heat exchanger
109. The heat rejected by the working substance to the regenerator in the Stirling cycle is:
a) Equal to the heat absorbed by the working substance from the regenerator
b) Not equal to the heat absorbed by the working substance from the regenerator
c) Dependent on the value of temperature in the heat addition process
d) Measured during the constant temperature heat rejection process

110. In a gas which flow into an expansion control and without doing work, this process is called:
a) Purging b) Scavenging c) **Throttling** d) Priming
111. The latent heat of vaporization decreases as the pressure and temperature of the liquid decreases. At the critical point, the heat of vaporization becomes what?
a) Increased b) **Zero** c) Decreased d) Constant
112. A fluid flowing in a reversible adiabatic deceleration process to zero velocity will reach a state known as:
a) Isentropic expansion b) Isentropic compression c) Adiabatic irreversibility d) **Isentropic Stagnation**
113. The ratio of the actual velocity to the sound velocity.
a) Prandtl Number b) Isotop number c) **Mach number** d) Torecelli's number
114. How do you call a refrigeration system in which only part of the refrigerant passing over the heat transfer separated from the vapor and re-circulated?
a) Absorption refrigeration system b) Cascade refrigeration system
c) **Flooded refrigeration system** d) Steam jet refrigeration system
115. What do you call a refrigeration system in which the refrigerant gas evolved in the evaporator is taken up in an absorber and released in a generator upon the application of heat?
a) **Absorption refrigeration system** b) Cascade refrigeration system
c) Flooded refrigeration system d) Steam jet refrigeration system
116. On the level of a molecule, the gas constant for one mole divided by the number of molecules in a mole is known as:
a) Avogadro's Number b) Planck Constant c) **Boltzmann's Constant** d) Gravitational Constant
117. Which of the following is true for white ice?
a) A fast-cooled water b) A sub-cooled water
c) **Due to dissolved air, gases and impurities** d) Formed by blowing air during freezing
118. In the detection of refrigerant leak, sulphur stick burning in the presence of ammonia will release of which of the following colors of smoke?
a) **Dense White** b) Dense Yellow c) Dense Orange d) Dense Red
119. What is a common cause of waterhammer in drying coils?
a) Wet steam supplied to the coil b) Too low a steam pressure onto the coil
c) **Condensate has to lift after the steam trap** d) The coil falling in the direction of steam flow
120. Which of the following statements is true?
a) Bimetallic steam traps are an ideal choice for rotating cylinders
b) Rotating cylinders can not suffer from steam locking
c) Strainers cannot be fitted to float traps which have a steam lock release
d) **Air vents around the thermodynamic and inverted bucket traps can considerably improve start-up times**
121. Name the principle cause of waterhammer?
a) Water particles suspended in steam b) **Water allowed to build up in pipes**
c) Water droplets carried along the insides of pipes d) Wet steam passing through steam traps
122. As a general rule, the cooling water side of the condenser should be scheduled for inspection every which of the following number of months?
a) Month b) **3 Months** c) 6 Months d) Year
123. How do you call a refrigeration system in which only part of the refrigerant passing over the heat transfer separated from the vapor and re-circulated?
a) Absorption refrigeration system b) Cascade refrigeration system
c) **Flooded refrigeration system** d) Steam jet refrigeration system
124. What do you call a refrigeration system in which the refrigerant gas evolved in the evaporator is taken up in an absorber and released in a generator upon the application of heat?
a) **Absorption refrigeration system** b) Cascade refrigeration system
c) Flooded refrigeration system d) Steam jet refrigeration system
125. What do you call a shut-off valve other than a valve for controlling the flow of refrigerant in a refrigeration system?
a) Fusible valve b) **Stop valve** c) Check valve d) Relief valve
126. Steam plant thermal efficiency may be increased by increasing the boiler pressure. However, the undesirable effect is that the moisture content will increase. This undesirable effect maybe2 corrected by;
a) exhausting to atmosphere b) **reheating** c) regenerative feed water installation d) decreasing condenser pressure
127. What is the working substance in jet refrigeration?
a) Ammonia b) R – 11 c) Brine & R – 12 d) **water**
128. What is a device, used in refrigeration system, having a predetermined temperature fusible member for the relief of pressures?
a) **Fusible valve** b) Stop valve c) Check valve d) Relief valve

129. Which of the following ranges of humidity ratio is used for comfort air conditioning?
a) 50 to 55 % **b) 55 to 60 %** c) 60 to 65 % d) 45 to 50 %
130. In cascade regenerative steam power plant. The device that allows liquid to be throttled to a lower region but remains vapor.
a) Throttle valve **b) Steam trap** c) Blow down valve d) Check valve
131. The production of more than one useful form of energy from the same energy source.
a) Thermodynamic process b) Reversible process **c) Regeneration** d) Combustion
132. When adding the refrigerant in a refrigeration system, it shall be charge into which of the following parts of the system?
a) High pressure side **b) Low pressure side** c) Low and high pressure side d) Compressor discharge line
133. Why is a boiler feed tank heated to approximately 85 °C?
a) To reduce the energy required to raise steam
b) To reduce the content of total dissolved solids in the water supplied to the boiler
c) To reduce the gas content of the water
d) To reduce the content of suspended solids in the water
134. Which of the following is the chemical formula for R – 12 or dichlorodifluoromethane?
a) CHCLF₂ b) **CCl₂F₂** c) CH₃Cl d) C₃H₈
135. Which of the following is not a desirable property of a refrigerant?
a) Low thermal conductivity b) Low freezing point c) Low condensing pressure d) Low viscosity
136. It is the process of treating air so as to control simultaneously its temperature, humidity, cleanliness and distribution to meet the comfort requirements of the occupants of the conditioned room or space. What is this?
a) Comfort Air Conditioning b) Summer Air Conditioning
c) Industrial Air Conditioning d) Commercial Air Conditioning
137. It is the ozone-destroying power of a substance measured relative to refrigerant 11 (R-11 or CFC-11).
a) Ozone Depletion Potential (ODP) b) Hydrochlorofluorocarbon (HCFC)
c) Hydrofluorocarbon (HFC) d) Global Warming Potential (GWP)
138. What always accompanies an isentropic expansion of steam?
a) An increase in entropy b) An increase in enthalpy c) A decrease in entropy **d) A decrease in enthalpy**
139. What is the advantage of a pressurized deaerator over an atmospheric deaerator?
a) A boiler feed tank is no longer required b) Less overall energy will be required to produce the steam
c) It can be fitted at ground level **d) It removes more oxygen**
140. At which typical pressure will a pressurized deaerator supplying a shell boiler normally operate?
a) 0.2 bar g b) 1.2 bar g c) 5 bar g d) Boiler pressure
141. How is the released oxygen in a pressurized deaerator prevented from being reabsorbed by the water?
a) By an air vent b) There is sufficient water surface for the air to be absorbed
c) By steam blanket over the water d) By the incoming steam against the incoming water
142. How is the water in a pressurized deaerator heated to the required temperature?
a) By a blanket of steam above the water b) By direct steam injection into the water
c) By a spray of steam as it enters the deaerator dome d) It is not heated further but just held at a higher pressure
143. In proximate analysis, the coal analysis is consisting of moisture content, fixed carbon, ash, and which of the following?
a) Sulfur b) Hydrogen c) Nitrogen **d) Volatile matter**
144. A relative measure of the ability of a substance to cause an increase in the temperature of the atmosphere by absorbing solar and earth radiation that is relative to the effect of refrigerant 11.
a) Ozone Depletion Potential (ODP) b) Hydrochlorofluorocarbon (HCFC)
c) Hydrofluorocarbon (HFC) **d) Global Warming Potential (GWP)**
145. A ratio calculated by dividing the cooling capacity in Btu/hr by the power input in watts and any set of rating conditions, expressed in Btu/Watt-hr; it is also defined as the amount of heat removed from the cooled space in Btu for one Watt-hour of electricity consumed.
a) Energy Efficiency Ratio (EER) b) Ozone Depletion Potential (ODP)
c) Global Warming Potential (GWP) d) Coefficient of Performance (COP)
146. Which of the following would cause low head pressure?
a) Too much cooling water b) Insufficient refrigeration gas
c) Insufficient cooling water **d) Too much cooling water and insufficient refrigeration gas**
147. What do you call the locus of elevations to which water will rise in the Piezometer tube?
a) Hydraulic gradient b) Space gradient c) Elevation gradient d) Velocity gradient
148. Which of the following is true if the dry-bulb and wet-bulb temperatures in air are identical?
a) The dew point of air is reached b) The mixture is completely dry
c) Relative humidity is zero d) Temperature of air cannot be determined
149. Which of the following defines Surging?

- a) A fixed mass flow rate irrespective of pressure ratio b) Change of mass flow rate in proportion to pressure ratio
c) **An unsteady, periodic and reversal of flow in the compressor** d) A reduction in lift force in the compressor
150. In a Brayton cycle or gas turbine cycle, what do you call the heat exchanger that passes exhaust heat to the compressed air, prior to combustion?
- a) Reheater **b) Recuperator** c) Heater d) Heat exchanger